

Comparing LLM and human reviews of social science research using data from Unjournal.org

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We will build and refine LLM tools to generate peer-reviews and ratings of impactful research, and compare these with human experts' work (esp. from Unjournal.org): to benchmark performance, understand AI's research taste, and develop tools to improve research evaluation and dissemination.

Table of contents

1	Introduction	4
2	Data and methods	5
2.1	Unjournal.org	5
2.2	LLM-based evaluation	5
2.2.1	Quantitative ratings	5
2.2.2	Journal ranking tiers	6
2.2.3	Claim identification	6
2.2.4	Qualitative assessments	6
3	Results	7
3.1	Quantitative ratings	7
3.1.1	Overall ratings	7
3.1.2	Individual metrics	9
3.1.3	Rationale behind the largest differences	9
3.2	Journal ranking tiers	9
3.3	Claim identification	9
3.4	Qualitative assessments	9
4	Discussion	12
4.1	Preliminary findings	12
4.2	Limitations	12
	References	13

1 Introduction

Work in progress

Pages, metrics, and comparisons are under active development. Expect rough edges and frequent updates.

Is AI good at peer-reviewing? Does it offer useful and valid feedback? Can it predict how human experts will rate research across a range of categories? How can it help academics do this “thankless” task better? Is it particularly good at spotting errors? Are there specific categories, e.g. spotting math errors or judging real-world relevance, where it does surprisingly well or poorly? How does its “research taste” compare to humans?

If AI research-evaluation works it could free up a lot of scientific resources – perhaps \$1.5 billion/year in the US alone (Aczel, Szaszi, and Holcombe (2021))) – and offer more continual and detailed review, helping improve research. It could also help characterize methodological strengths/weaknesses across papers, aiding training and research direction-setting. Furthermore, a key promise of AI is to directly improve science and research. Understanding how AI engages with research evaluations may provide a window into its values, abilities, and limitations.

In this project, we are testing the capabilities of current large language models (LLMs), illustrating whether they can generate research paper evaluations comparable to expert human reviews. The Unjournal systematically prioritizes ‘impactful’ research and pays for high-quality human evaluations, structured quantified ratings, claim identification and assessment, and predictions. In this project, we use an AI (OpenAI’s GPT-5 model for now) to review social science research papers under the same criteria used by human reviewers for The Unjournal.

Each paper is assessed on specific dimensions – for example, the strength of its evidence, rigor of methods, clarity of communication, openness/reproducibility, relevance to global priorities, and overall quality. The LLM will provide quantitative scores (with uncertainty intervals) on these criteria and produce a written evaluation

Our initial dataset will include research papers that have existing human evaluations. For each paper, the AI will generate: (1) numeric ratings on the defined criteria, (2) identification of the paper’s key claims, and (3) a detailed review discussing the paper’s contributions and weaknesses. We will then compare the AI-generated evaluations to the published human evaluations.

Next, we will focus on papers currently under evaluation, i.e., where no human evaluation has been made public, to allow us to rule out any contamination.

2 Data and methods

We draw on two main sources:

- 1) Human evaluations from The Unjournal’s public dataset (PubMed reports and the Coda evaluation form export).
- 2) LLM-generated evaluations produced by GPT-5 using a structured JSON-schema prompt.

We report (i) the human dataset and curation, and (ii) the LLM evaluation pipeline (PDF ingestion → prompt → JSON result → tidy CSV). Code cells below show the exact configuration and contract used with the model, as well as the API call.

2.1 Unjournal.org

[Unjournal Dat Blog](#)

2.2 LLM-based evaluation

2.2.1 Quantitative ratings

Following The Unjournal’s [standard guidelines for evaluators](#) and their [academic evaluation form](#), evaluators are asked to consider each paper along the following dimensions: **claims & evidence**, **methods**, **logic & communication**, **open science**, **global relevance**, and an **overall** assessment. Ratings are interpreted as percentiles relative to serious recent work in the same area. For each metric, evaluators are asked to provide a midpoint and a 90% credible interval to communicate uncertainty.

Response schema & system prompt: This section defines the *contract* we expect back from the model:

- **METRICS** lists the categories we score.
- **metric_schema** specifies the shape and bounds of each metric.
- **response_format** (JSON Schema) asks the model to return only a JSON object conforming to that schema.
- **SYSTEM_PROMPT** explains scoring semantics (percentiles + 90% credible intervals) and how to set the **overall** score.

Below is the main helper that runs a single evaluation:

- extracts text with `pdf_to_string()`,

- calls Chat Completions with the strict JSON Schema in `response_format`,
- parses the returned JSON string into a Python dict.

Batch-evaluate all PDFs: This loop walks every `*.pdf` in `papers/`, calls `evaluate_paper()`, pads a 0.8s sleep for rate-limits, and collects the raw JSON dicts in memory (`records`). The nested `records` structure is converted to a tidy long format and written to `results/metrics_long.csv`.

2.2.2 Journal ranking tiers

- What journal ranking tier should this work be published in? (0.0-5.0)
- What journal ranking tier will this work be published in? (0.0-5.0)

2.2.3 Claim identification

2.2.4 Qualitative assessments

3 Results

Here we present preliminary results, starting with a comparison of the LLM-generated quantitative ratings (, see the [previous section](#)) with human evaluations across [the Unjournal’s criteria](#).

Results on journal ranking tiers, claim identification, and full written reports will be added soon.

3.1 Quantitative ratings

3.1.1 Overall ratings

We begin by comparing *overall* scores.

In Figure 3.2, the orange diamond and line show the midpoint and its 90% credible interval for each paper, while the green circles show every individual human evaluator’s midpoint (with 90% CI when provided).

Figure 3.1: Overall ratings (horizontal jitter). Human (green) vs LLM (orange). Hollow dot marks the median per rater.

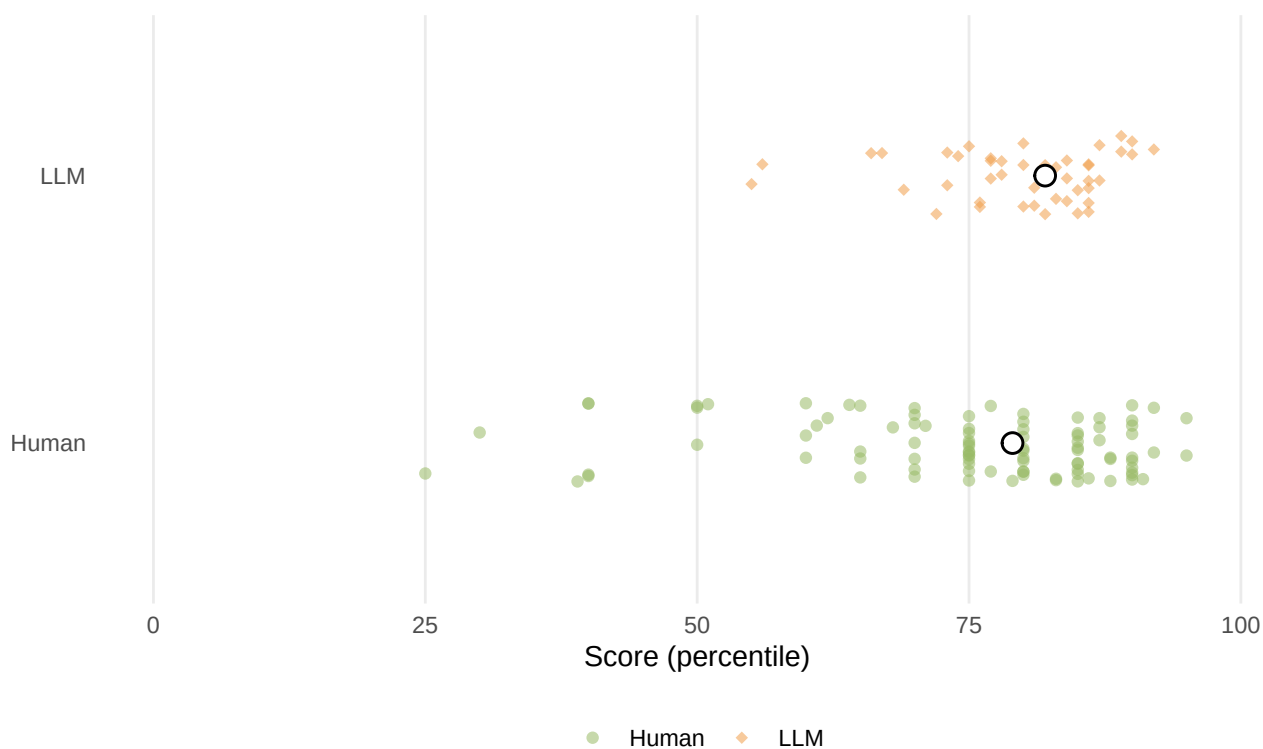
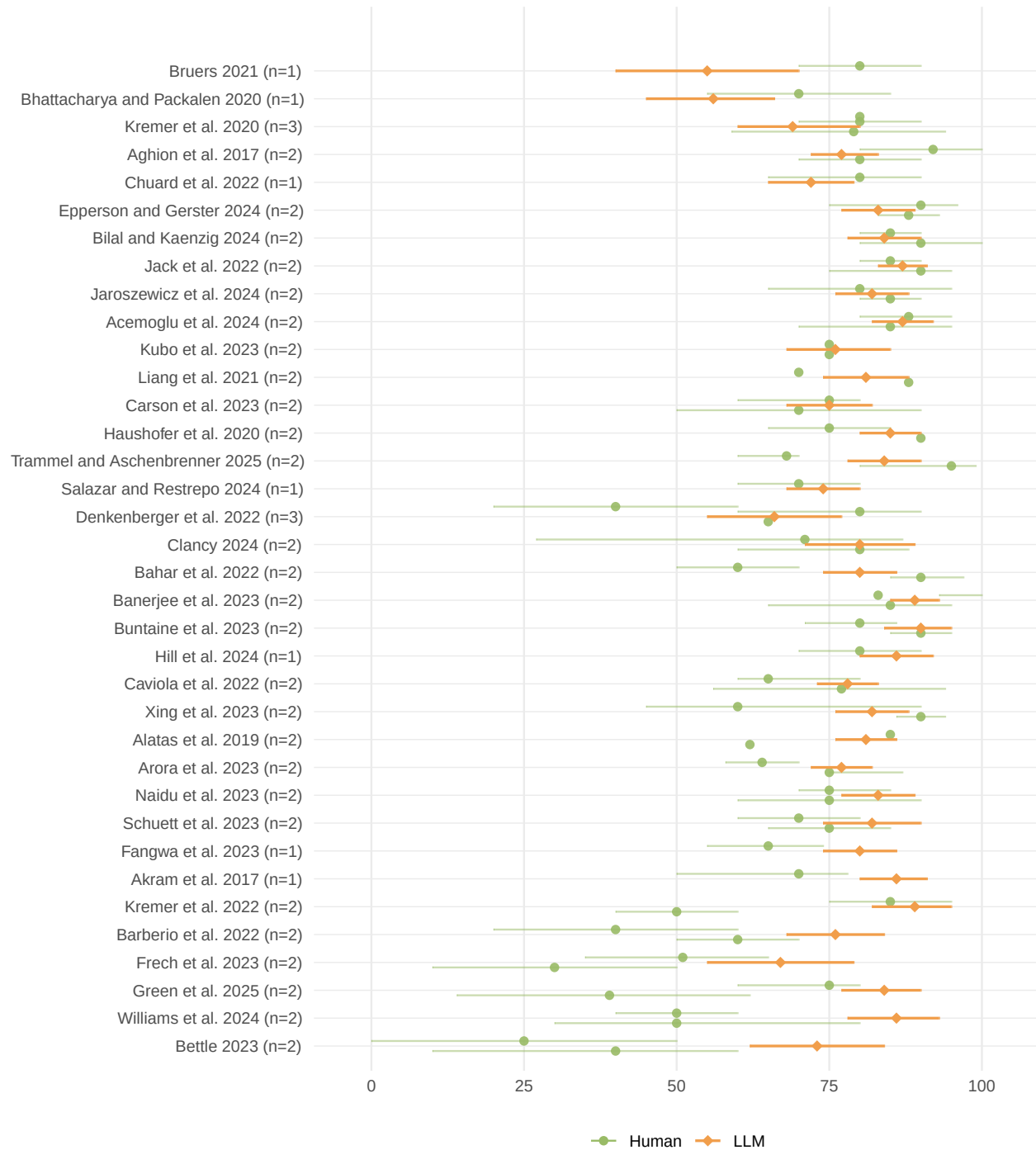


Figure 3.2: LLM vs individual Human ratings (overall). Each rating on its own lane: orange LLM midpoint+CI centered; green human midpoints (thin CI when available) offset per rater.



Next, Figure 3.3 contrasts relative *ranks* of papers under human and LLM scoring. Lines curve upward when the LLM ranks a paper higher than humans (orange), downward when lower (green).

3.1.2 Individual metrics

We then look closer at the specific ratings on the criterion level. Figure 3.4 displays LLM – Human differences by paper $\times$ metric; orange cells indicate higher LLM scores, green cells lower ones.

Table 3.1: Overall agreement across all metrics

criteria	n	pearson	spearman	kappa_unwtd	kappa_quad
adv_knowledge	35	0.348	0.476	0.058	0.228
claims	13	0.151	0.336	0.103	0.287
gp_relevance	36	0.314	0.173	0.186	0.130
logic_comms	36	-0.010	0.116	0.007	0.002
methods	35	0.210	0.427	0.097	0.327
open_sci	36	0.135	0.084	0.055	0.096
overall	36	0.289	0.392	0.119	0.327

3.1.3 Rationale behind the largest differences

As a first shot at understanding sources of disagreement, we look at the reported rationale for those ratings that diverge the most from (average) human assessments.

3.2 Journal ranking tiers

3.3 Claim identification

3.4 Qualitative assessments

Figure 3.3: Relative ranking (overall) by LLM and Human evaluators

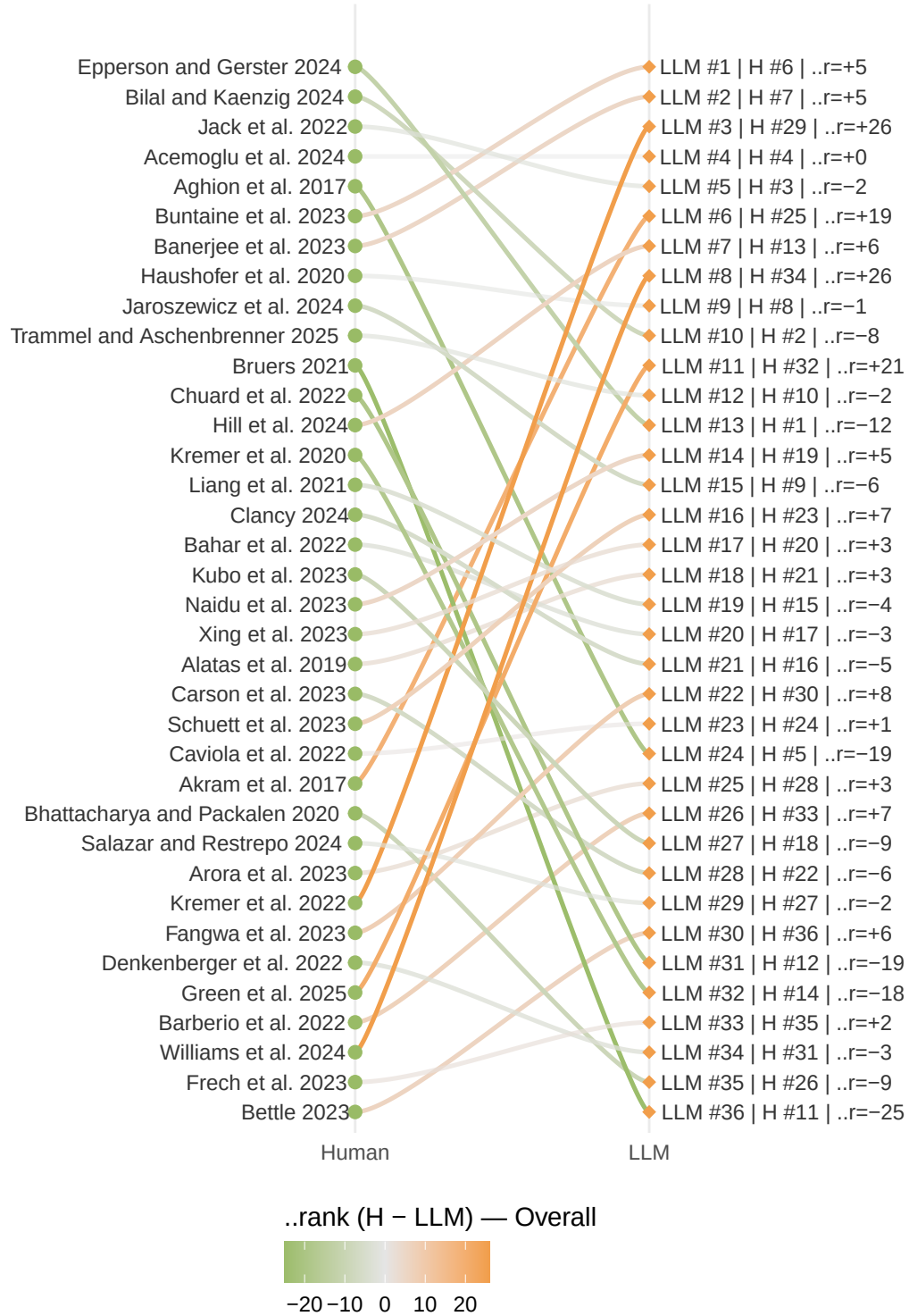
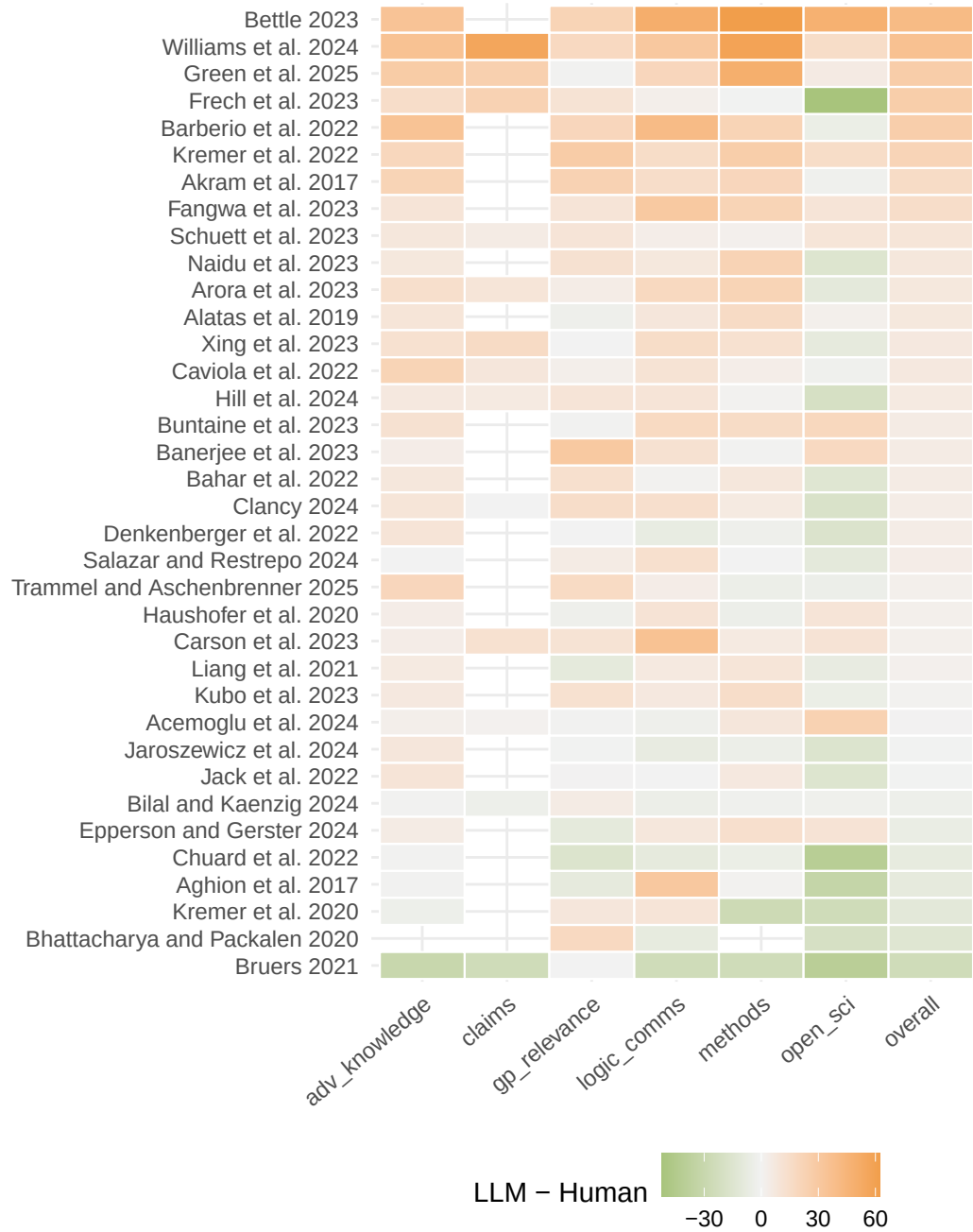


Figure 3.4: LLM – Human differences by paper × metric



4 Discussion

4.1 Preliminary findings

4.2 Limitations

Planned updates and extesions

- Journal ranking tiers and predictions
- Claim identification
- Qualitative assessments and full evaluations
- Improved PDF ingestion
- Optimized System prompt
- Alternative models
- Extended set of papers

References

We used R v. 4.4.2 (R Core Team 2024) and the following R packages: ggforce v. 0.5.0 (Pedersen 2025), here v. 1.0.1 (Müller 2020), janitor v. 2.2.1 (Firke 2024), knitr v. 1.50 (Xie 2014, 2015, 2025), renv v. 1.1.5 (Ushey and Wickham 2025), rmarkdown v. 2.29 (Xie, Allaire, and Golemund 2018; Xie, Dervieux, and Riederer 2020; Allaire et al. 2024), stringi v. 1.8.7 (Gagolewski 2022), tidyverse v. 2.0.0 (Wickham et al. 2019).

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